

1. SYNOPSIS

The Drone Detection System for Security in the Airforce is an advanced AI-powered surveillance framework designed to detect and classify drones using computer vision techniques. The system involves the collection and training of a dataset comprising various drone types, including Multi-Rotor Drones, Fixed-Wing Drones, Single-Rotor Drones, Hybrid VTOL Drones, Nano and Micro Drones, and AI-Powered Drones. It employs the Canny Edge Detection Algorithm to efficiently detect drones in real-time and utilizes a Convolutional Neural Network (CNN) to classify the detected drones. Upon successful classification, the system plays an audio announcement of the identified drone type, ensuring immediate awareness for security personnel.

Once a drone is detected, the system automatically captures a 10-second video footage using OpenCV for further analysis and evidence collection. Additionally, the system integrates an SMTP-based email alert mechanism to notify the in-charge officer with the classification details and the captured footage, enabling prompt security responses. This project enhances airforce base security by leveraging AI-driven automation, reducing manual monitoring efforts, and improving real-time threat detection capabilities. The integration of deep learning, computer vision, and automated alerting makes this system an essential asset for military surveillance and defense applications.

2 SYSTEM REQUIREMENTS

The system requirements includes the hardware and software requirements that helps to run the application. System requirements outline the hardware and software specification necessary for a particular software application or system to operate effectively. These requirements ensure that users have the necessary infrastructure in place to run the software smoothly and without issue. By understanding and meeting the system requirements outlined for a software application, users can ensure that they have the necessary hardware and software infrastructure in place to run the software effectively and achieve optimal performance.

HARDWARE REQUIREMENTS

Processor : Intel Dual Core

Processor RAM : 4 GB

Hard Disk : 500 GB

SOFTWARE REQUIREMENTS

- Front End : Python IDE 3.7.1, OpenCV
- Back End : TensorFlow

3. SYSTEM DESCRIPTION

WINDOWS 11

Windows 11 is the latest major release of Microsoft's Windows NT operating system. It is the successor to Windows 10, which had been released five years earlier. Windows 11 was officially announced on June 24, 2021, and became generally available on October 5, 2021. The operating system introduces a refreshed design and several new features aimed at improving user experience, performance, and security. It is available as a free upgrade for eligible Windows 10 devices via Windows Update. One of the most notable changes in Windows 11 is its redesigned user interface, featuring a centered Start menu and taskbar, rounded corners, and revamped system icons, contributing to a more modern and streamlined look. The operating system also introduces features like Snap Layouts and Snap Groups, which improve multitasking and window management, and virtual desktops have been enhanced with more customization options. In terms of hardware requirements, Windows 11 has stricter minimum system requirements than its predecessor. It requires a 64-bit processor with at least 1 GHz clock speed and two or more cores, 4 GB of RAM, and 64 GB of storage. Additionally, Windows 11 mandates the presence of TPM (Trusted Platform Module) 2.0, a secure boot process with UEFI firmware, and a DirectX 12-compatible graphics card for the best experience. The introduction of these requirements has meant that some older devices are not eligible for an upgrade to Windows 11.

Windows 11 places a stronger emphasis on integration with Microsoft's cloud services, including OneDrive, Microsoft 365, and Azure Active Directory, enabling enhanced productivity and collaboration. Furthermore, One of the major improvements introduced in Windows 11 is its redesigned graphical user interface. The operating system features a centered Start Menu, redesigned taskbar, rounded window corners, updated icons, improved animations, and an overall cleaner visual appearance. These enhancements provide a more organized and user-friendly workspace, allowing users to navigate applications and system settings with greater ease. The Snap Layouts and Snap Groups features significantly improve multitasking by enabling users to arrange multiple application windows efficiently. This functionality is particularly beneficial for software developers who frequently work with code editors, terminals, web browsers, database management tools, and documentation

Python

Python is a high-level, general-purpose and a very popular programming language. Python programming language (latest Python 3) is being used in web development, Machine Learning applications, along with all cutting edge technology in Software Industry. Python Programming Language is very well suited for Beginners, also for experienced programmers with other programming languages like C++ and Java.

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

- **Python is Interpreted** – Python is processed at runtime by the interpreter. User do not need to compile ymy program before executing it. This is similar to PERL and PHP.
- **Python is Interactive** – User can actually sit at a Python prompt and interact with the interpreter directly to write ymy programs.
- **Python is Object-Oriented** – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.
- **Python is a Beginner's Language** – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Python Programming Language

1. Python is currently the most widely used multi-purpose, high-level programming language.
2. Python allows programming in Object-Oriented and Procedural paradigms.
3. Python programs generally are smaller than other programming languages like Java. Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.
4. Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.
5. The biggest strength of Python is huge collection of standard library which can be used for the following:
 - Machine Learning
 - GUI Applications (like Kivy, Tkinter, PyQtetc.)
 - Web frameworks like Django (used by YouTube, Instagram, Dropbox)
 - Image processing (like OpenCV, Pillow)
 - Web scraping (like Scrapy, BeautifulSoup, Selenium)
 - Test frameworks
 - Multimedia
 - Scientific computing
 - Text processing

Python Features

Python's features include –

- **Easy-to-learn** – Python has few keywords, simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- **Easy-to-read** – Python code is more clearly defined and visible to the eyes.
- **Easy-to-maintain** – Python's source code is fairly easy-to-maintain.
- **A broad standard library** – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- **Interactive Mode** – Python has support for an interactive mode which allows interactive testing and debugging of snippets of code.
- **Portable** – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- **Extendable** – User can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- **Databases** – Python provides interfaces to all major commercial databases.
- **GUI Programming** – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- **Scalable** – Python provides a better structure and support for large programs than shell scripting.

PACKAGES

TensorFlow

Image recognition consists of pixel and pattern matching to identify the image and its parts. Each image is a container of pixels which in turn are the combination of numbers. These numbers represent the color depth. Here the character recognition can be executed by this TensorFlow framework.

Scikit-learn

It provides a selection of efficient tools for machine learning and statistical modeling including classification, regression, clustering and dimensionality reduction via a consistency interface in Python.

Tkinter

Tk widgets can be used to construct buttons, menus, data fields, etc. in a Python application. Here the application is designed with button to upload the input image.

NumPy

NumPy can be used to perform a wide variety of mathematical operations on arrays. The comparison and prediction with the dataset can be implemented by using various mathematical calculation in recognizing the handwritten character input.

Matplotlib

Matplotlib is a cross-platform, data visualization and graphical plotting library for Python and its numerical extension NumPy.

Pillow

Pillow is a Python Imaging Library (PIL), which adds support for opening, manipulating, and saving images.

ipython

It is an interactive command-line terminal for Python. IPython offers an enhanced read-eval-print loop (REPL) environment particularly well adapted to scientific computing.

SpeechPy

SpeechPy is an open source Python package that contains speech preprocessing techniques, speech features, and important post-processing operations. It provides most frequent used speech features including MFCCs and filter bank energies alongside with the log-energy of filter-banks. The aim of the package is to provide projector's with a simple tool for speech feature extraction and processing purposes in applications such as Automatic Speech Recognition and Speaker Verification.

NEURAL NETWORKS

Neural network is a system inspired by human brain function, consists of neurons connected in parallel with the ability to learn. A basic design of neural network has input layer, hidden layer, and output layer. Neural networks, with their remarkable ability to derive meaning from complicated or imprecise data, can be used to extract pattern and detect trends that are too complex to be noticed by either humans or other computer techniques.

A trained neural network can be thought of as an “expert” in the category of information it has been to analyze. Neural networks are particularly useful for solving problems that cannot be expressed as a series of steps, such as recognizing patterns, classifying them into groups, series prediction and data mining.

NETWORK LAYERS

The commonest type of artificial neural network consists of three groups, or layers, of units: a layer of “input” units is connected to a layer of “hidden” units, which is connected to a layer of “output” units.

- The activity of the input units represents the raw information that is fed into the network.
- The activity of each hidden unit is determined by the activities of the input units and the weights on the connections between the input and the hidden units.
- The behavior of the output units depends on the activity of the hidden units and the weights between the hidden and output units.

This simple type of network is interesting because the hidden units are free to construct their own representations of the input. The weights between the input and the hidden units determine when each hidden unit is active, and so by modifying these weights, a hidden unit can choose what it represents.

Tensor Flow

TensorFlow can train and run deep neural networks for handwritten digit classification, image recognition, word embeddings, recurrent neural networks, sequence-to-sequence models for machine translation and natural language processing

TensorFlow working

TensorFlow allows developers to create dataflow graphs—structures that describe how data moves through a graph, or a series of processing nodes. Each node in the graph represents a mathematical operation, and each connection or edge between nodes is a multidimensional data array, or tensor. TensorFlow provides all of this for the programmer by way of the Python language. Python is easy to learn and work with, and provides convenient ways to express how high-level abstractions can be coupled together. Nodes and tensors in TensorFlow are Python objects, and TensorFlow applications are themselves Python applications. The actual math operations, however, are not performed in Python. The libraries of transformations that are available through TensorFlow are written as high-performance C++ binaries. Python just directs traffic between the pieces, and provides high-level programming abstractions to hook them together.

TensorFlow applications can be run on most any target that's convenient: a local machine, a cluster in the cloud, iOS and Android devices, CPUs or GPUs. If you use Google's own cloud, you can run TensorFlow on Google's custom TensorFlow Processing Unit (TPU) silicon for further acceleration. The resulting models created by TensorFlow, though, can be deployed on most any device where they will be used to serve predictions. TensorFlow 2.0, released in October 2019, revamped the framework in many ways based on user feedback, to make it easier to work with (e.g., by using the relatively simple [Keras](#) API for model training) and more performant.

Distributed training is easier to run thanks to a new API, and support for TensorFlow Lite makes it possible to deploy models on a greater variety of platforms. However, code written for earlier versions of TensorFlow must be rewritten—sometimes only slightly, sometimes significantly—to take maximum advantage of new TensorFlow 2.0 features.

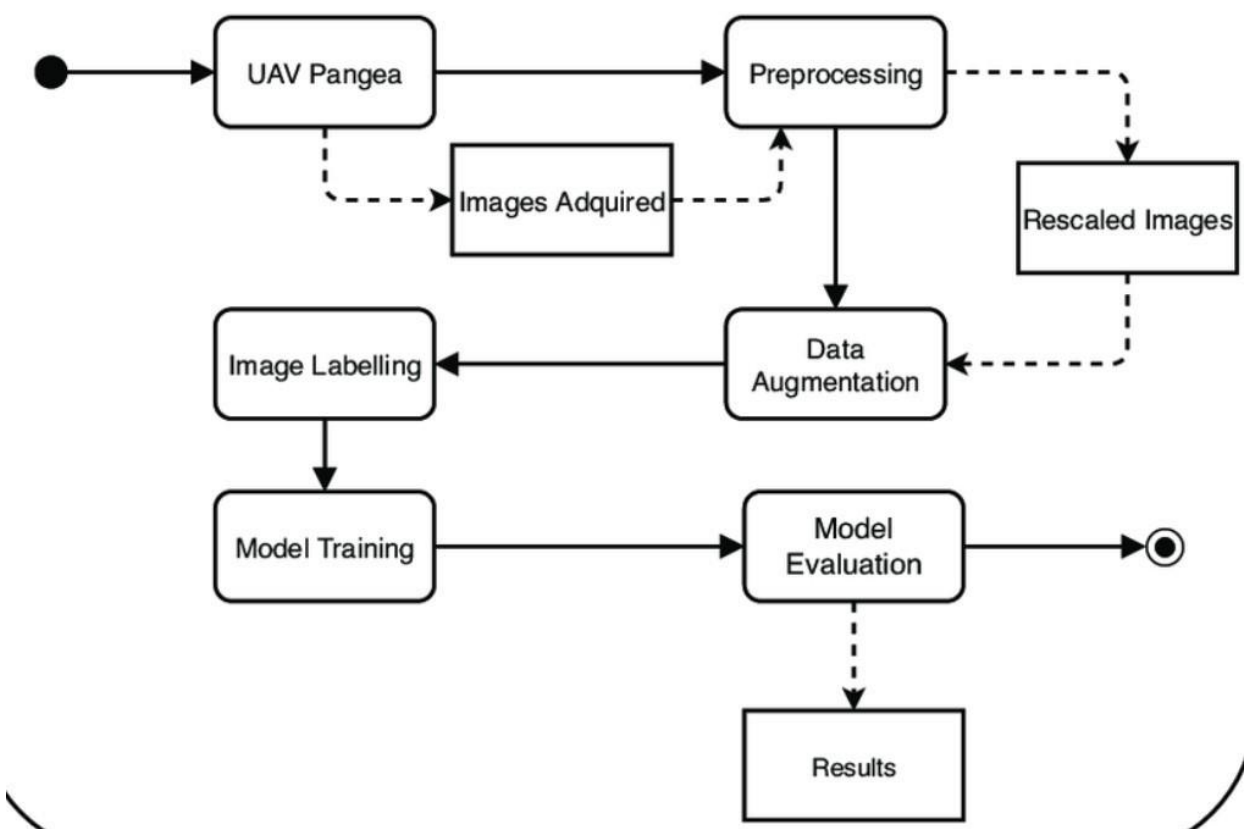
TensorFlow benefits

TensorFlow offers additional conveniences for developers who need to debug and gain introspection into TensorFlow apps. The eager execution mode lets you evaluate and modify each graph operation separately and transparently, instead of constructing the entire graph as a single opaque object and evaluating it all at once. The Tensor Board visualization suite lets you inspect and profile the way graphs run by way of an interactive, web-based dashboard.

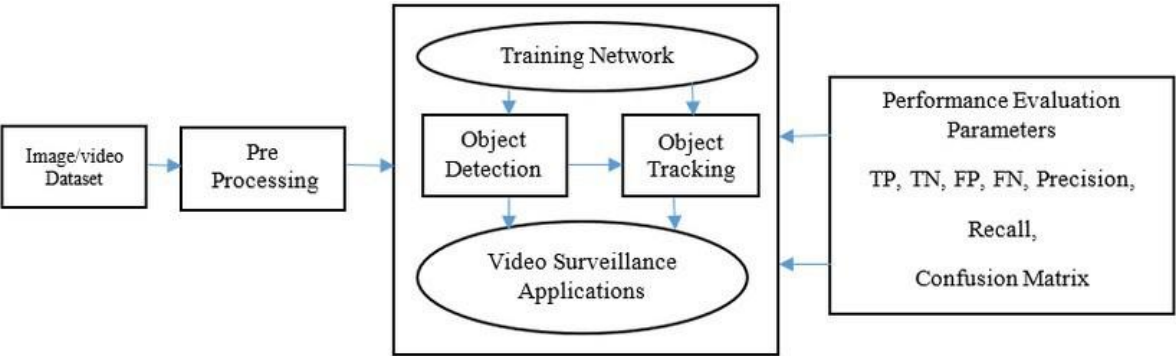
Tensor Flow also gains many advantages from the backing of an A-list commercial outfit in Google. Google has not only fueled the rapid pace of development behind the project, but created many significant offerings around TensorFlow that make it easier to deploy and easier to use: the above-mentioned TPU silicon for accelerated performance in Google's cloud; an online hub for sharing models created with the framework; in-browser and mobile-friendly incarnations of the framework; and much more.

4. SYSTEM DESIGN

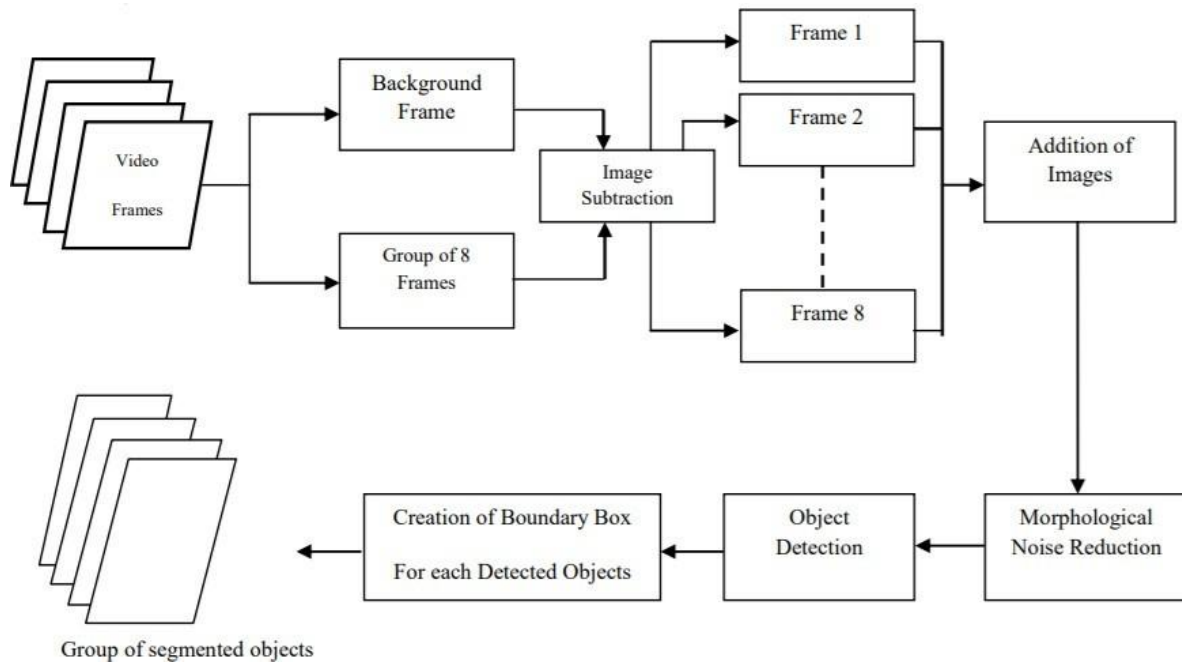
4.1 Data Flow Diagram



Work Flow in Drone Detection



Analyze and Classify Drone Type Prediction



About Algorithm

Convolutional Neural Networks

CNN are made up of a large number of interconnected neurons that have learnable weights and biases. In the architecture of CNN the neurons are organized as layers. It consists of an input layer, many hidden layers and an output layer. If the network has a large number of hidden layers the same are generally referred to as deep neural networks. The neurons in the hidden layers of CNN are connected to a small region (receptive field) of the input space generated from the previous layer instead of connecting to all, as in the fully connected network like Multi Layered Perceptron (MLP) networks. This approach reduces the number of connection weights (parameters) in CNN compared to MLP. As a consequence, CNN takes less time to train for the networks of similar size. The input to the typical CNN is a two-dimensional (2D) array of data such as images. Unlike the regular neural network the layers of a CNN are arranged in three dimensions (width, height and depth).

The followings are the type of layers which are commonly found in the CNNs.

- Input Layer is basically a buffer to hold the input and pass it on to the next layer.

- Convolution Layer performs the core operation of feature extraction. It performs Convolution operation of the input data. The convolution operation is executed by sliding the kernel over the input, and performs sum of the product at every location. The step size with which the kernel slides are known as stride. Numerous convolution operations are performed on the input by using different kernel, which results in different feature maps, the number of feature map produced in a convolutional layer is also known as the depth of the layer.
- Rectified Linear Unit (ReLU) is an activation function used to introduce non linearity. It replaces negative value with zero, which can speed up the learning process. The output of every convolution layer is passed through the activation function.
- Pooling layer reduces the spatial size of each feature map, which in turn reduces the computation in the network. Pooling also uses a sliding window that moves in stride across the feature map and transform it into representative values. Min pooling, average pooling and max pooling are commonly used.
- Fully connected layer connects every neuron in the layer to all the neurons in the previous layer. It learns the non-linear combination of the features and used for classifying or predicting the output. For classification problems, the fully connected layer is generally followed by a soft-max layer, it produces the probability of each class for the given input. And for regression problems, it is followed by a regression layer to predict the output.

The architecture of the CNN is organized according to the problem to be addressed. Like any other neural networks the CNN is intelligent and they can learn. Learning is achieved through training (supervised). The CNN are feedforward networks and uses back-propagation training algorithm. The training is performed in two passes; forward and backward pass. In the forward pass the network weight and bias are initialized with small random numbers and compute the network output by using training input. The error is computed by comparing the network output with the desired training output. In the backward pass the error propagates backward and all the weights and bias are adjusted to minimize the error. The process is repeated until the desired result is obtained. Once the network is trained with a suitable dataset, it can be used for solving a specific problem.

Many researches have developed different CNN architectures for classification. Some of the most popular deep CNN networks include AlexNet, ZfNet and LeNet.

AlexNet

AlexNet accepts input of size $227 \times 227 \times 3$ and consist of 25 layers. It has 5 convolution layers which are followed by ReLU layer and max pooling layer. The first two convolution layers are also followed by crosschannel normalization layer. Cross channel normalization carries out channel wise normalization. It replaces each element with a normalized value obtained from neighbouring cells. It also has three fully connected layers followed by ReLU Layer and the first two fully connected layer is also followed by dropout layers. The output of the last fully connected layer is given as an input to softmax which produces the probability distribution of 1000 classes. The AlexNet was designed for recognizing objects in ImageNet

ZfNet

ZfNet has similar architecture to AlexNet but differs in the filter size and the number of filters used. In the first convolution layer, the AlexNet uses filter of size 11×11 with a stride of 4 and the ZfNet uses 7×7 and stride of 2. The ZfNet uses 512,1024 and 512 filters in the third,fourth and fifth convolution layer whereas the AlexNet uses 384,384 and 256 filters respectively. The ZfNet was also designed for recognizing objects in ImageNet.

LeNet

LeNet accepts a grayscale image of size 32×32 as an input. The architecture consists of two sets of convolutional and average pooling layers, followed by a flattening convolutional layer, then follows two fully connected layers and a softmax layer for classification. The LeNet is a pioneering work in recognizing hand written digits from the images by using CNN.

Dataset

<https://www.kaggle.com/datasets/dasmehdixtr/drone-dataset-uav>

Drone Dataset (UAV)

▲ 272

<> Code

Data Card

Code (14)

Discussion (4)

Suggestions (0)

About this directory

Suggest Edits

Datasets for both yolo and tensorflow

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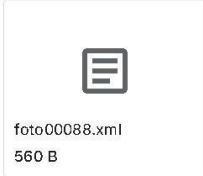
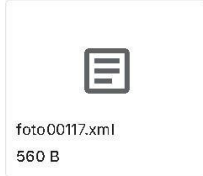
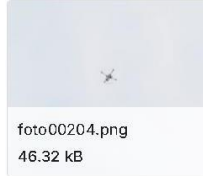
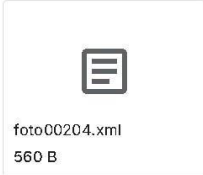
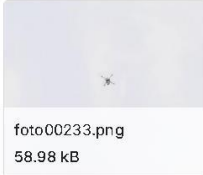
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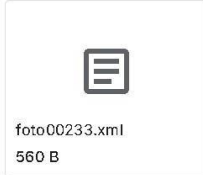
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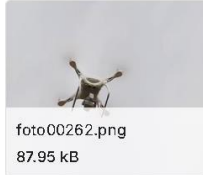
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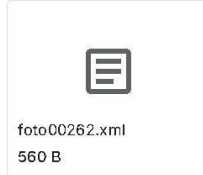
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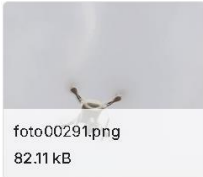
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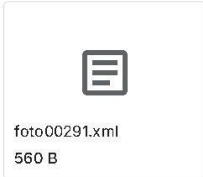
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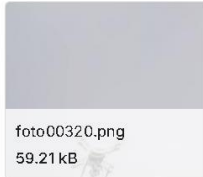
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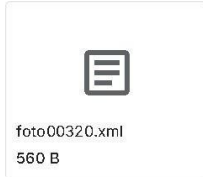
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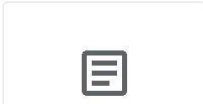
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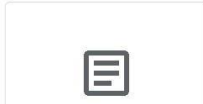
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The **Drone Dataset (UAV)**, curated by Mehdi Özel and hosted on Kaggle, is a comprehensive collection of images focusing on rotary-wing unmanned aerial vehicles (UAVs). This dataset was assembled for a UAV competition and addresses the scarcity of publicly available drone image datasets suitable for training machine learning models.

Dataset Composition

The dataset comprises 4,913 image files, totaling approximately 751 MB in size. These images capture various rotary-wing UAVs from multiple perspectives and under different conditions, providing a diverse set of scenarios for model training and evaluation.

Annotations and Metadata

Each image in the dataset is accompanied by label files that provide detailed annotations. These annotations include information about the presence and position of drones within the images, facilitating tasks such as object detection and classification. The structured labeling enhances the dataset's utility for developing and testing computer vision algorithms aimed at drone detection.

Applications and Use Cases

This dataset is particularly valuable for researchers and developers working on drone detection systems, especially those utilizing algorithms like YOLO (You Only Look Once) for real-time object detection. Its comprehensive nature allows for the training of robust models capable of identifying drones in various environments, contributing to advancements in security, surveillance, and autonomous navigation systems.

Access and Licensing

The Drone Dataset (UAV) is publicly accessible through Kaggle and can be downloaded for research and development purposes. Users are encouraged to review the dataset's licensing terms on Kaggle to ensure compliance with usage guidelines.

Once the classification is complete, the system triggers an audio alert announcing the detected drone type to notify the security team immediately. Simultaneously, it records a 10-second video of the detected drone using OpenCV for further analysis and evidence collection. The system also integrates an SMTP-based email alert mechanism, which sends an automated email to the in-charge officer with the classification details and recorded footage. By automating drone detection, classification, and alerting, this project enhances airforce security, ensuring quick response to potential threats while minimizing manual monitoring efforts. The integration of AI, real-time detection, and automated alerts makes this system a powerful tool for military surveillance and defense applications.

5.SYSTEM IMPLEMENTATION

Data Collection and Preprocessing Module

The Data Collection and Preprocessing Module is responsible for gathering a comprehensive dataset of different drone types, including Multi-Rotor, Fixed-Wing, Single-Rotor, Hybrid VTOL, Nano/Micro, and AI-Powered Drones. The dataset is sourced from open-source repositories, drone manufacturers, and real-world drone images captured using high-resolution cameras. To ensure accurate classification, the dataset must contain a variety of drone orientations, lighting conditions, and backgrounds. Additionally, images are labeled according to their respective drone categories, creating a structured dataset for training the deep learning model.

Once the dataset is collected, it undergoes a preprocessing phase to enhance the robustness of the Convolutional Neural Network (CNN) model. This includes resizing images to a standard input size, normalizing pixel values to improve learning efficiency, and applying data augmentation techniques such as rotation, flipping, and contrast adjustment. These preprocessing steps ensure that the model can accurately detect and classify drones in various environmental conditions, improving real-time detection reliability.

Real-Time Drone Detection Module

The Real-Time Drone Detection Module is responsible for capturing video footage from high-resolution surveillance cameras or dedicated drone detection sensors. The system continuously processes the live feed to detect the presence of a drone in restricted airspace. To enhance detection accuracy, the module employs the Canny Edge Detection algorithm, which highlights the edges and contours of objects in a frame. This technique helps differentiate drones from background noise, birds, and other flying objects, making it ideal for real-time monitoring.

Once an object with drone-like characteristics is detected, the module isolates the region of interest (ROI) and sends it for further classification. The Canny Edge Detection technique ensures faster and more precise detection, even in challenging environments such as low visibility, nighttime conditions, and high-altitude surveillance. By continuously monitoring airspace for unauthorized drones, this module acts as the first

line of defense in the security system.

Deep Learning-Based Drone Classification Module

The Deep Learning-Based Drone Classification Module uses a Convolutional Neural Network (CNN) to accurately classify detected drones into one of the six predefined categories: Multi-Rotor, Fixed-Wing, Single-Rotor, Hybrid VTOL, Nano/Micro, and AI-Powered Drones. The CNN model extracts high-level features such as shape, size, rotor configuration, and texture patterns to differentiate between various drone types. The deep learning model is trained on an extensive dataset, ensuring high accuracy and robustness.

When a new drone is detected, the system processes its features through multiple CNN layers, including convolution, pooling, and fully connected layers, to generate a classification output. The classified drone type is immediately displayed on the user interface, allowing security personnel to assess the situation. By using a highly optimized CNN model, this module ensures rapid and precise classification, reducing false alarms and improving the overall effectiveness of the airforce security system.

Audio Alert Module

The Audio Alert Module enhances security response time by immediately triggering an automated voice announcement upon successful drone classification. Once the deep learning model identifies the drone type, the system converts the text output into a speech-based alert, which is played through security speakers or communication devices. This instant audio alert ensures that security personnel are notified in real time, allowing them to take swift action.

The text-to-speech (TTS) engine is integrated into this module to dynamically generate speech outputs for different drone types. For example, if a Multi-Rotor Drone is detected, the system will play "Alert! Multi-Rotor Drone Detected!". This module can be customized to include additional security commands, language preferences, and volume controls to adapt to different operational environments. By providing hands-free notifications, the Audio Alert Module improves situational awareness and rapid threat response.

Video Recording Module

The Video Recording Module ensures that all drone detection incidents are documented through a 10-second real-time video clip. Once a drone is detected, the module automatically triggers OpenCV to start recording the live footage. The video is stored in a secure database or local storage system, allowing security teams to review the footage for further analysis. This feature is particularly useful for tracking repeated intrusions, gathering intelligence on drone behavior, and providing evidence for security reports.

To optimize storage management, the module implements automatic overwriting and categorization features, ensuring that only relevant drone footage is stored. The recorded videos are named using a timestamp-based format and can be accessed via the system's user interface. By providing visual confirmation of drone activity, the Video Recording Module enhances security monitoring and investigation capabilities.

Email Notification Module

The Email Notification Module ensures that security personnel are alerted immediately upon drone detection through automated email notifications. This module integrates SMTP (Simple Mail Transfer Protocol) to send an email to the in-charge officer, containing critical details such as drone classification, timestamp, and a link to the recorded 10-second footage. This ensures that security teams can assess the threat remotely and coordinate appropriate responses.

To prevent spam or unnecessary alerts, the module prioritizes detections based on risk levels. For example, a Nano/Micro Drone may trigger only a standard alert, while an AI-Powered Drone could initiate a high-priority security warning. The email format includes structured data with embedded images and video attachments, providing a comprehensive situational overview. By enabling real-time communication between the system and security personnel, the Email Notification Module significantly improves threat response efficiency and documentation.

User Interface Live

Drone Detection Panel

The Live Drone Detection Panel serves as the core interface for monitoring

aerial security threats in real time. It provides a live video streaming window from surveillance cameras, ensuring continuous monitoring of airspace. This module employs Canny Edge Detection to highlight drone contours, enabling security personnel to distinguish drones from background objects like birds or other flying objects. The detection results are displayed with an outlined bounding box around the identified drone, improving visual clarity for the monitoring team.

Additionally, this module provides a detection confidence score (%), which represents the certainty of the detection model in identifying the drone. The interface includes Capture Image and Record Video buttons, allowing users to manually trigger snapshot captures or initiate recording beyond the automated 10-second video clip. These options are particularly useful in manual verification scenarios, where operators may want to gather additional visual evidence of drone activity.

Drone Classification Section

The Drone Classification Section is responsible for categorizing the detected drone using a Convolutional Neural Network (CNN)-based classifier. Once a drone is detected, the CNN model analyzes the image and classifies it into one of the predefined categories: Multi-Rotor, Fixed-Wing, Single-Rotor, Hybrid VTOL, Nano/Micro, or AI-Powered Drones. The classification results are presented with an image preview of the detected drone, along with the confidence percentage to indicate model accuracy.

A real-time classification log is maintained, displaying the detected drone category, timestamp, and confidence score. This ensures that security personnel can track and review previous detections efficiently. Moreover, the system provides an option for manual verification, allowing security teams to cross-check the classification results and provide feedback for model improvement. This feedback loop enhances the system's learning capability, ensuring higher classification accuracy over time.

Audio Alert Control Panel

The Audio Alert Control Panel is designed to immediately notify security personnel upon the detection of an unauthorized drone. It generates automated voice announcements corresponding to the detected drone type, providing real-time alerts without requiring manual intervention. The system uses pre-recorded voice messages

to announce classifications such as *"Multi-Rotor Drone Detected"* or *"Fixed-Wing Drone Detected"*, ensuring a swift security response.

Additionally, this module allows users to toggle the voice alert ON or OFF based on operational requirements. Security officers can also adjust the alert volume and trigger a manual announcement for specific cases. This ensures that alerts can be customized based on environmental noise levels and urgency. The presence of custom alert options enhances system usability, making it adaptable for different security scenarios, such as training exercises or real-world threat detections.

Video Recording & Playback Module

The Video Recording & Playback Module plays a crucial role in documentation and post-incident analysis. When a drone is detected, the system automatically records a 10-second video clip using OpenCV, capturing crucial footage for investigation. The recorded videos are stored in a designated secure folder, ensuring availability for future reference. This helps security teams analyze drone behavior, flight patterns, and possible security breaches.

The Playback Interface allows security officers to review recorded footage directly from the system dashboard. Users can play, pause, rewind, or fast-forward recordings for detailed examination. The module also offers Download and Share options, enabling officers to export footage for forensic analysis or share it with higher authorities.

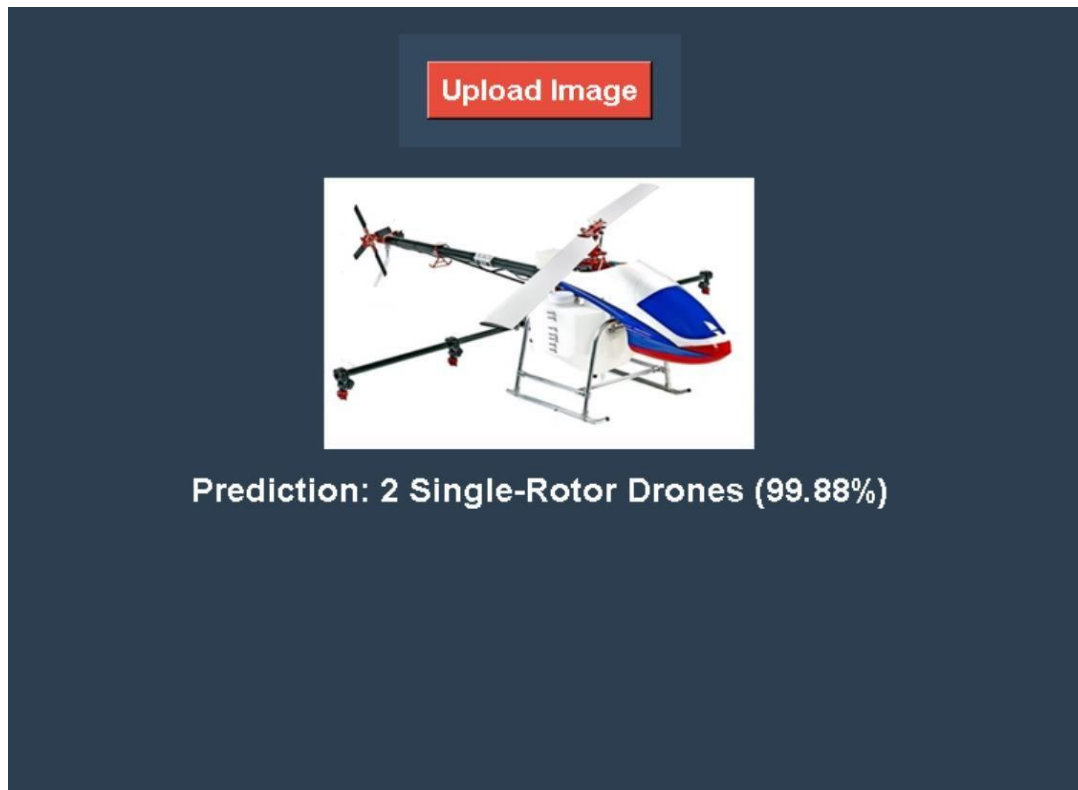
Email Notification Center

The Email Notification Center is responsible for sending automated security alerts to designated in-charge officers whenever an unauthorized drone is detected.

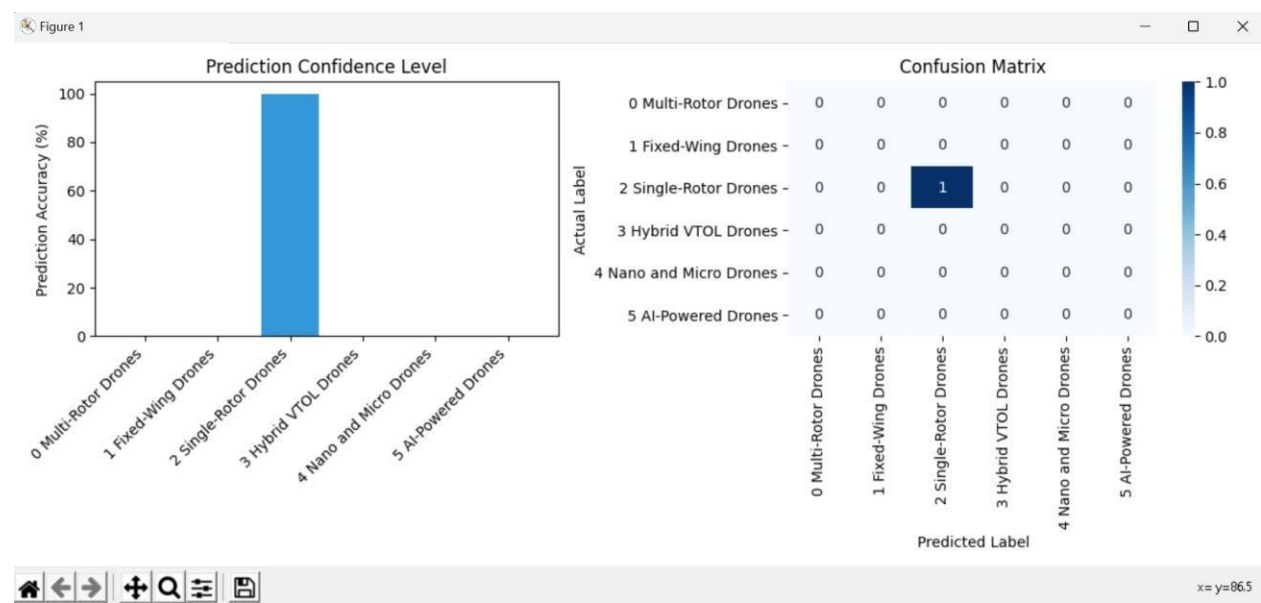
A detailed log of sent emails is maintained within the module, displaying timestamps and recipient details for verification purposes. Users can preview email content before sending, ensuring accuracy in notifications. Additionally, security officers can add or edit recipient email addresses, enabling flexibility in communication management. In case of an email failure, a Resend option is provided, ensuring that critical alerts are always delivered without delay. This feature enhances real-time threat response by ensuring timely communication of security breaches.

5.1 SCREENSHOTS

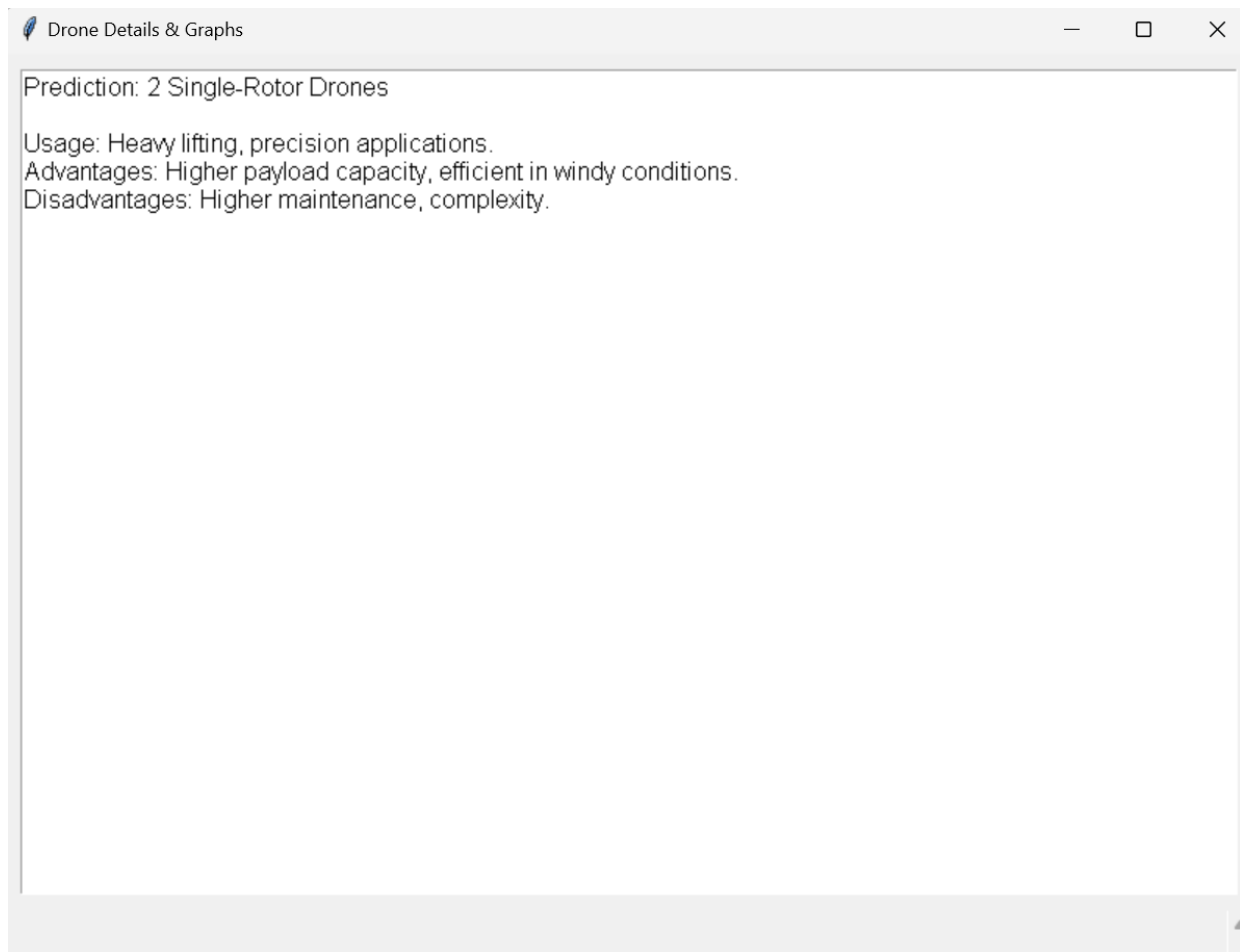
5.1.2 Upload Drone Image



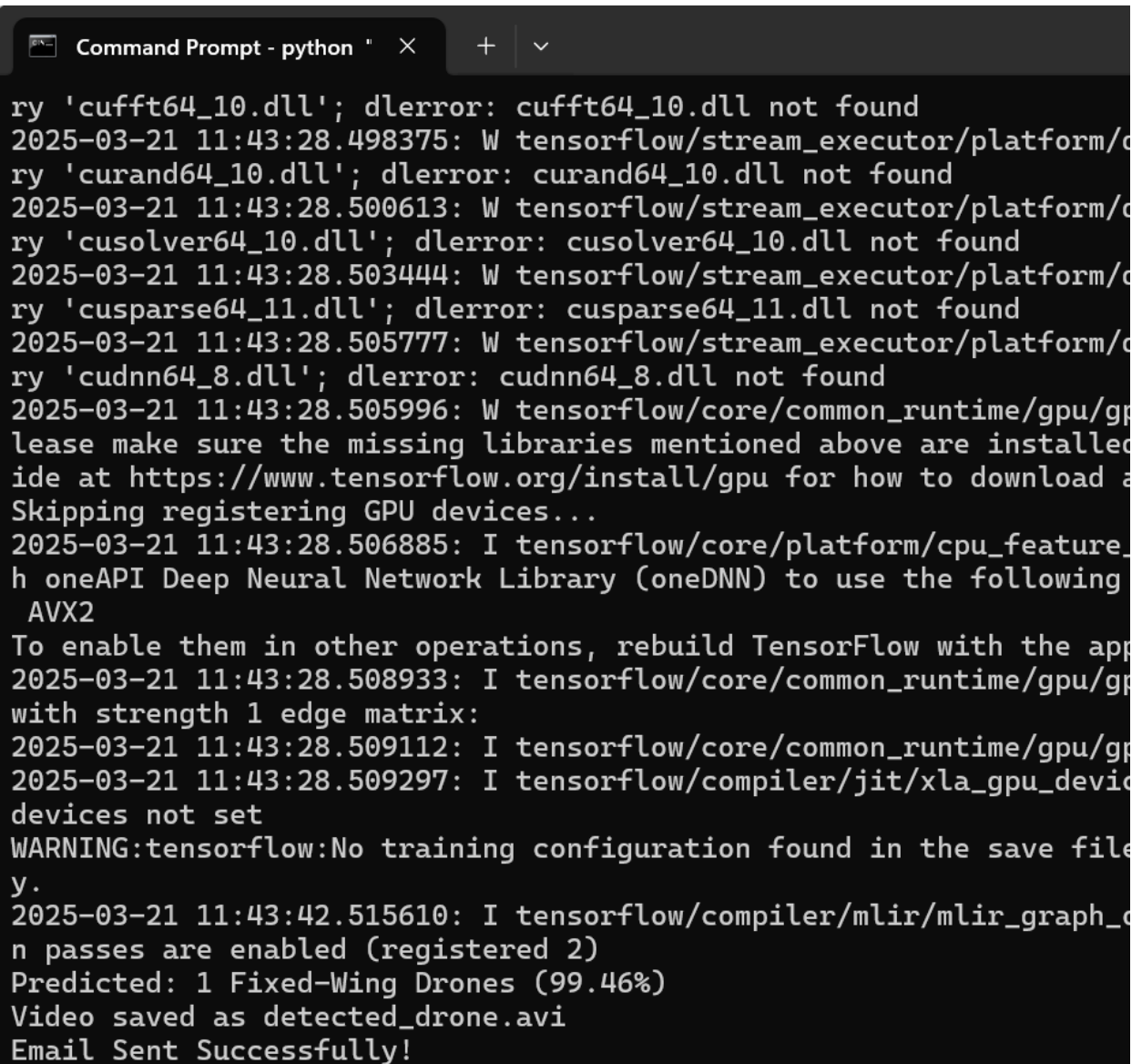
5.1.3 Prediction Result



5.1.4 Predicted Output

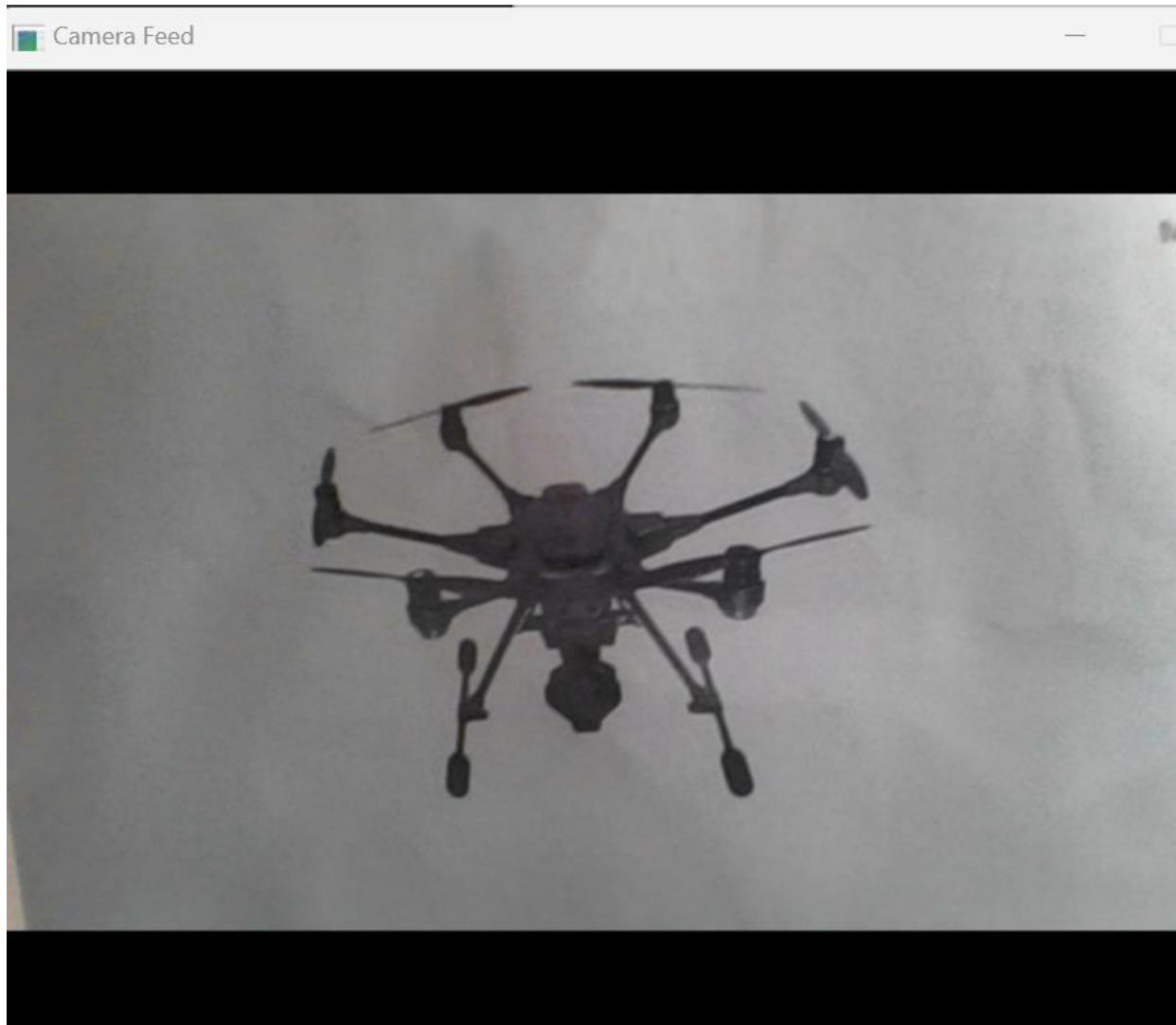


5.1.5 Run Drone Prediction

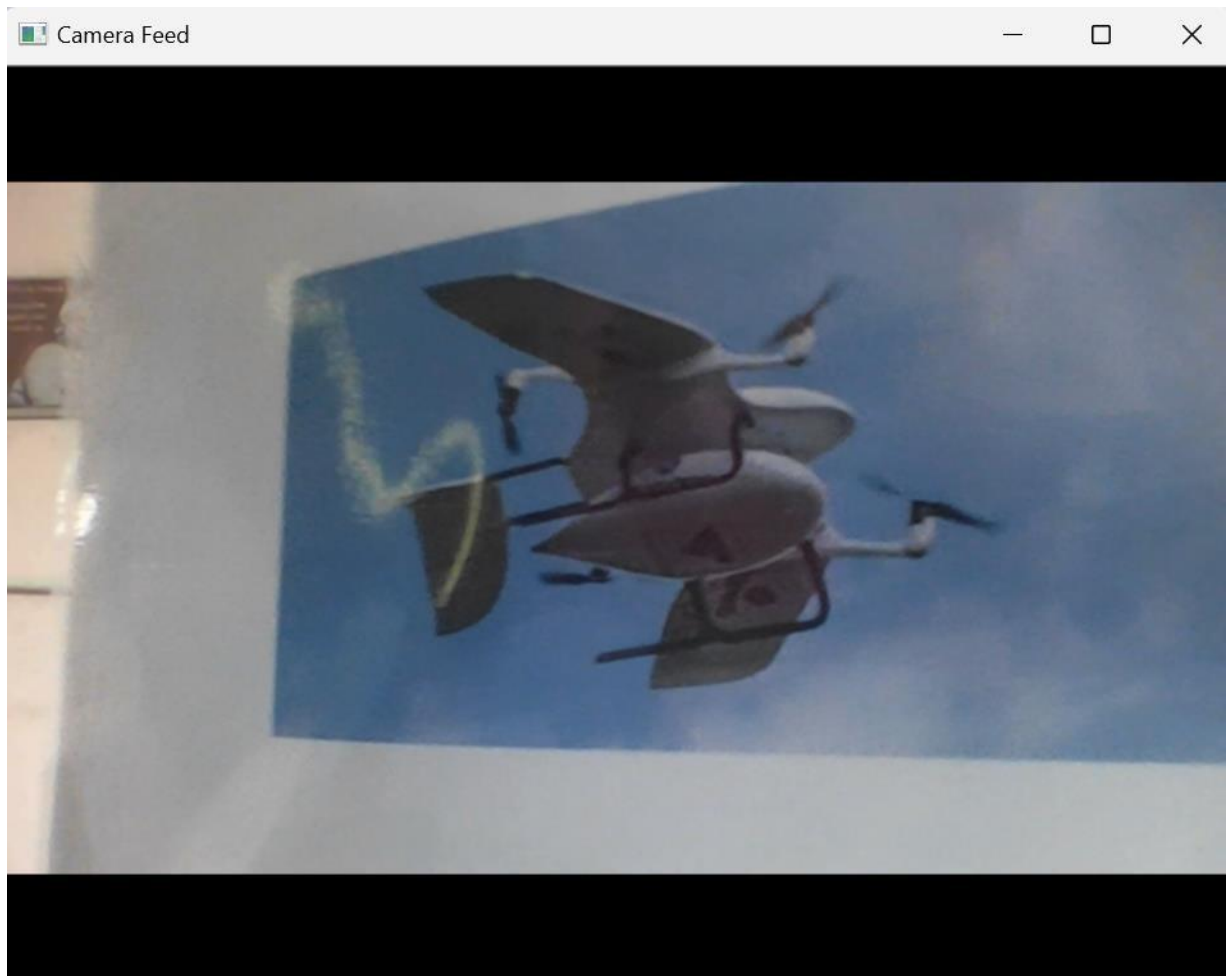


```
Command Prompt - python
ry 'cuffft64_10.dll'; dLError: cuffft64_10.dll not found
2025-03-21 11:43:28.498375: W tensorflow/stream_executor/platform/c
ry 'curand64_10.dll'; dLError: curand64_10.dll not found
2025-03-21 11:43:28.500613: W tensorflow/stream_executor/platform/c
ry 'cusolver64_10.dll'; dLError: cusolver64_10.dll not found
2025-03-21 11:43:28.503444: W tensorflow/stream_executor/platform/c
ry 'cusparse64_11.dll'; dLError: cusparse64_11.dll not found
2025-03-21 11:43:28.505777: W tensorflow/stream_executor/platform/c
ry 'cudnn64_8.dll'; dLError: cudnn64_8.dll not found
2025-03-21 11:43:28.505996: W tensorflow/core/common_runtime/gpu/gp
lease make sure the missing libraries mentioned above are installed
ide at https://www.tensorflow.org/install/gpu for how to download a
Skipping registering GPU devices...
2025-03-21 11:43:28.506885: I tensorflow/core/platform/cpu_feature_
h oneAPI Deep Neural Network Library (oneDNN) to use the following
AVX2
To enable them in other operations, rebuild TensorFlow with the app
2025-03-21 11:43:28.508933: I tensorflow/core/common_runtime/gpu/gp
with strength 1 edge matrix:
2025-03-21 11:43:28.509112: I tensorflow/core/common_runtime/gpu/gp
2025-03-21 11:43:28.509297: I tensorflow/compiler/jit/xla_gpu_devic
devices not set
WARNING:tensorflow:No training configuration found in the save file
y.
2025-03-21 11:43:42.515610: I tensorflow/compiler/mlir/mlir_graph_c
n passes are enabled (registered 2)
Predicted: 1 Fixed-Wing Drones (99.46%)
Video saved as detected_drone.avi
Email Sent Successfully!
```

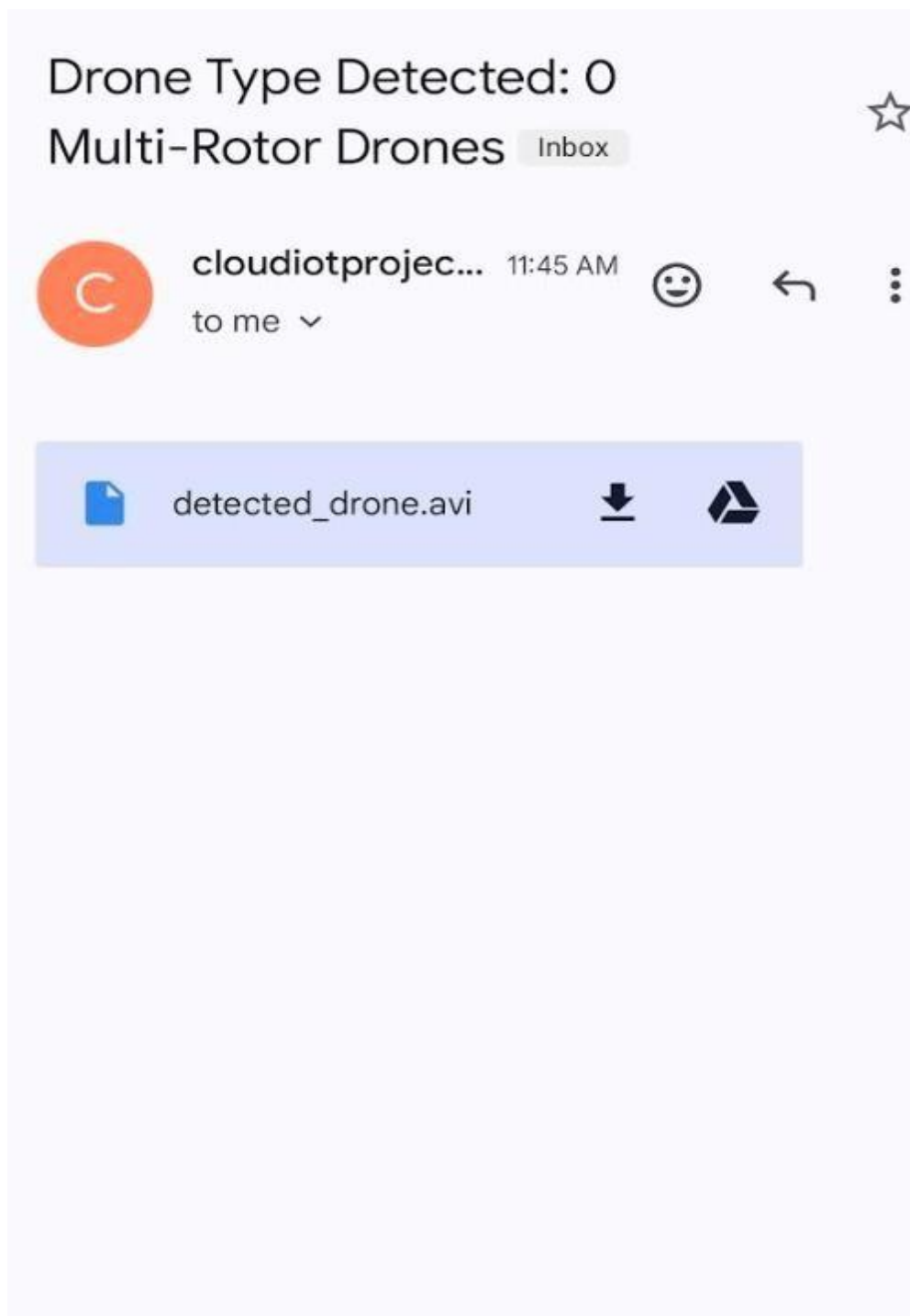
5.1.6 Drone Detection via Camera



5.1.7 Capture Video Footage



5.1.8 Video Alert



Conclusion

The Drone Detection System is a powerful and innovative security solution designed to enhance airforce surveillance by accurately detecting, classifying, and monitoring drones in real time. By integrating Canny Edge Detection for contour identification and CNN-based classification for drone type recognition, the system ensures a high level of precision in distinguishing between different drone categories, including Multi-Rotor, Fixed-Wing, Single-Rotor, Hybrid VTOL, Nano/Micro, and AI-Powered Drones. The incorporation of automated audio alerts, real-time video recording, and email notifications enhances situational awareness, allowing security personnel to respond swiftly to potential threats. This multi-layered approach strengthens airforce security by reducing unauthorized drone intrusions and improving operational efficiency.

While the system performs effectively under controlled conditions, it can be further enhanced through advanced AI techniques, edge computing, and thermal imaging integration to improve detection in low-light and adverse weather conditions. Additionally, faster deep learning models like YOLO and real-time anomaly detection can further refine its accuracy and response time. Overall, this project lays the foundation for a scalable, AI-driven drone security system, which can be expanded for broader military, commercial, and civilian applications, ensuring enhanced surveillance and airspace protection in the future.

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